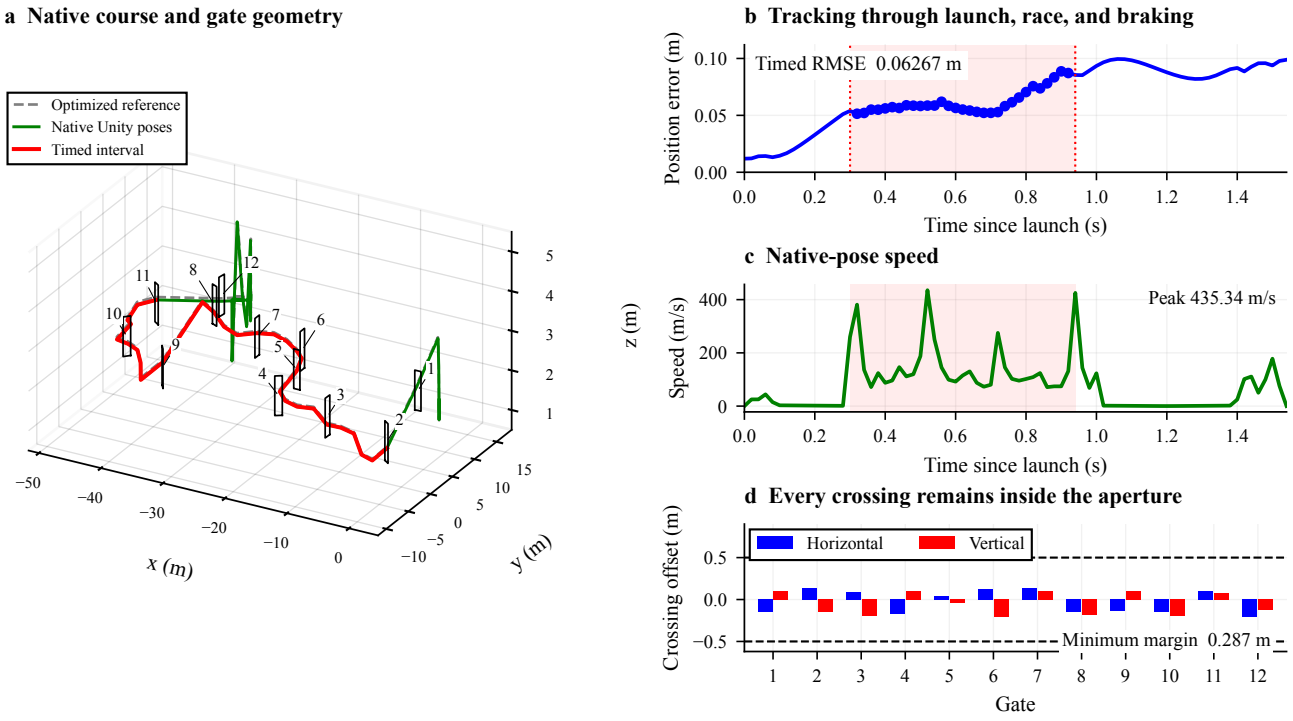

a Native course and gate geometry
Optimized reference
Native Unity poses
Timed interval
x (m)
y (m)
z (m)
b Tracking through launch, race, and braking
Timed RMSE 0.06267 m
Position error (m)
Time since launch (s)
c Native-pose speed
Peak 435.34 m/s
Speed (m/s)
Time since launch (s)
d Every crossing remains inside the aperture
Horizontal
Vertical
Minimum margin 0.287 m
Crossing offset (m)
Gate